Lecture Slides: Photosynthesis*Live*Test

SLIDE 1: Title - Photosynthesis Fundamentals

Subtitle: How plants convert solar radiation into biological chemical energy.

Key themes: Solar energy capture, Chloroplast ultrastructure, Calvin cycle biochemical pathways.

SLIDE 2: LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

• Formulate and explain the balanced chemical equation for oxygenic photosynthesis.

• Identify and describe the structural functions of thylakoid membranes, grana, and stroma.

• Trace the sequence of electron flow through photosystem II and photosystem I in the Z-scheme.

• Summarize the three major phases of the Calvin cycle occurring in the stroma.

SLIDE 3: TOPIC INTRODUCTION & CONTEXT

• Photosynthesis is the fundamental biological reaction sustaining nearly all life on Earth.

• Autotrophic organisms capture electromagnetic radiation and convert it into chemical bond energy.

• The overall balanced reaction consumes 6 carbon dioxide and 6 water molecules to yield glucose and oxygen.

• Terrestrial plants, aquatic algae, and cyanobacteria act as the primary photosynthetic producers.

• The released oxygen gas maintains Earth's atmospheric composition necessary for aerobic respiration.

Speaker note: Emphasize that cyanobacteria were the original oxygenators of early Earth.

SLIDE 4: CORE CONCEPT 1: CHLOROPLAST STRUCTURE & THYLAKOIDS

• Photosynthesis takes place inside specialized double-membrane organelles termed chloroplasts.

• The stroma is a dense, protein-rich fluid matrix containing enzymes required for carbohydrate synthesis.

• Thylakoids are flattened, disc-shaped membranous sacs organized into stacks known as grana.

• Chlorophyll a and accessory pigments are embedded directly into the thylakoid lipid bilayer.

• The lumen inside thylakoids maintains a concentrated proton gradient during photochemical activity.

Speaker note: Have students note the structural similarity between chloroplasts and cyanobacteria.